Problem Set 11

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: MML 6.1–6.5 (probability), 8.1 and 8.3.1 (maximum likelihood), 9.1–9.2.1 (maximum likelihood for linear regression); class notes on probability (expected utility). MML is Deisenroth et al. (2021).

1 Computing Probabilities

Consider a family with three children, each of whom is either a boy (B) or a girl (G). The sample space $\Omega := \{1, \ldots, 8\}$ consists of the eight outcomes listed in the table below. The random variables $C_1, C_2, C_3 \colon \Omega \to \{\mathrm{B}, \mathrm{G}\}$ represent the first, second, and third child, respectively.

Ω	C_1	C_2	C_3
1	B	B	B
2	B	B	G
3	B	G	B
4	B	G	G
5	G	B	B
6	G	B	G
7	G	G	B
8	G	G	G

Each of the outcomes is equally probable: $P(\omega) = \frac{1}{8}$ for each $\omega \in \Omega$.

Answer the following questions. Restate the question *formally* as a probabilistic statement, if applicable, and then compute its value.

1. What is the probability of the first child being a girl?
2. What is the probability of the second child being a boy if the first child is a girl?
3. What is the probability of the second child being a boy if at least one of the children is a boy?
4. Use *Bayes' theorem* to compute the probability that the first child is a boy if at least one of the children is a boy.
5. Are the events of the first child being a boy and the second child being a girl independent?
6. What is the average number of girls in a family?
7. Construct a probability mass function for the random variable C_3.
8. Let $N \colon \Omega \to \{0, 1, 2, 3\}$ be the number of girls in the family. Construct a table of the joint probability mass function $\mathbb{P}[C_1 = c, N = n]$ for $c \in \{\mathrm{B}, \mathrm{G}\}$ and $n \in \{0, 1, 2, 3\}$. Then compute the marginal probability mass function p_N by summing the joint probability mass function over the values of C_1.

	$n = 0$	$n = 1$	$n = 2$	$n = 3$
$\mathbb{P}[C_1 = \mathrm{B}, N = n]$				
$\mathbb{P}[C_1 = \mathrm{G}, N = n]$				
$\mathbb{P}[N = n]$				

2 A Sibling Conundrum

Consider the following question:

> Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?

1. Formalize the problem posed above to get and answer of $\frac{1}{3}$.
2. Formalize the problem posed above to get and answer of $\frac{1}{2}$.

3 Maximizing Expected Utility

Imagine you are given two identical envelopes, each containing money. One contains twice as much as the other. You may pick one envelope and keep the money it contains. Having chosen an envelope at will, but before inspecting it, you are given the chance to switch envelopes. Should you switch?

1. Let X be the random variable that represents the amount of money in the chosen envelope. How much do you expect to gain from switching the envelopes?
2. Does it mean you are always better off switching the envelopes? Explain.

4 Maximum Likelihood for a Coin

Suppose that you toss a coin m times and want to estimate the probability $\theta \in [0, 1]$ that it comes up heads. We model the tosses as independent random variables $Y_1, \ldots, Y_m$ such that $\mathbb{P}[Y_i = 1] = \theta$ and $\mathbb{P}[Y_i = 0] = 1 - \theta$, where $Y_i = 1$ when the i-th toss is heads. Let $y_1, \ldots, y_m \in \{0, 1\}$ be the observed outcomes and let $k := \sum_{i=1}^{m} y_i$ be the number of heads.

1. Show that the likelihood of the observed outcomes is $L(\theta) = \theta^k (1 - \theta)^{m-k}$, and write the log-likelihood $\ell(\theta) := \log L(\theta)$.
2. Assume that $0 < k < m$. Derive the maximum likelihood estimate $\theta^\star$ by maximizing $\ell(\theta)$, and verify that it is indeed a maximum.
3. What is $\theta^\star$ when $m = 3$ and all three tosses come up heads? Do you think that this is a reasonable estimate? Explain.

5 Maximum Likelihood for Linear Regression

Suppose that you are given m data points with features $\boldsymbol{a}_i \in \mathbb{R}^n$ and responses $y_i \in \mathbb{R}$ for $i = 1, \ldots, m$. We model the responses as independent random variables $Y_1, \ldots, Y_m$ such that Y_i is normally distributed with mean $\boldsymbol{a}_i^\top \boldsymbol{x}$ and variance σ^2, where $\boldsymbol{x} \in \mathbb{R}^n$ and $\sigma > 0$ are unknown parameters. That is, each Y_i has the probability density function

$$p_{Y_i}(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y - \boldsymbol{a}_i^\top \boldsymbol{x})^2}{2\sigma^2}\right), \qquad y \in \mathbb{R}.$$

Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be the matrix with rows $\boldsymbol{a}_1^\top, \ldots, \boldsymbol{a}_m^\top$ and assume that its columns are linearly independent, and let $\boldsymbol{b} := (y_1, \ldots, y_m) \in \mathbb{R}^m$.

1. Write the log-likelihood $\ell(\boldsymbol{x}, \sigma)$ of the observed responses $y_1, \ldots, y_m$ as a function of the parameters $\boldsymbol{x}$ and σ.
2. Show that the maximum likelihood estimate $\boldsymbol{x}^\star$ minimizes $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2$; that is, maximum likelihood estimation with normally distributed noise is the same as least squares. Give a closed-form expression for $\boldsymbol{x}^\star$.
3. Derive the maximum likelihood estimate of σ^2.

6 Maximum Likelihood for Log-Normal Regression

Many quantities, such as incomes or house prices, are positive and right-skewed, and are therefore commonly modeled using a log-normal distribution. Repeat Problem 5 but assume that Y_i is log-normally distributed with the probability density function

$$p_{Y_i}(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log y - \boldsymbol{a}_i^\top \boldsymbol{x})^2}{2\sigma^2}\right), \qquad y > 0.$$

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be the matrix with rows $\boldsymbol{a}_1^\top, \ldots, \boldsymbol{a}_m^\top$ and assume that its columns are linearly independent.

1. Write the log-likelihood $\ell(\boldsymbol{x}, \sigma)$ of the observed responses $y_1, \ldots, y_m$ as a function of the parameters $\boldsymbol{x}$ and σ.
2. Using your results from Problem 5, show that the maximum likelihood estimate $\boldsymbol{x}^\star$ minimizes $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2$ where $\boldsymbol{b} \in \mathbb{R}^m$ is the vector with $b_i = \log y_i$, and give a closed-form expression for $\boldsymbol{x}^\star$. Explain why the factor $1/y$ in the density does not affect $\boldsymbol{x}^\star$.
3. Derive the maximum likelihood estimate of σ^2.
4. Show that $\mathbb{E}[Y_i] = \exp(\boldsymbol{a}_i^\top \boldsymbol{x} + \sigma^2/2)$.
5. Then explain why predicting the response of a new data point $\boldsymbol{a}$ as $\exp(\boldsymbol{a}^\top \boldsymbol{x}^\star)$ tends to underestimate its mean.